The Prime Triplet as a Three-Member Two-Centre CRT Pattern: A Non-Monotone ω-Distortion on the $6N$ Skeleton

Ruqing Chen
GUT Geoservice Inc., Montreal, Quebec, Canada
ruqing@hotmail.com

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Abstract

Parts IX–X showed that prime pairs straddling two consecutive centres of the $6N$ skeleton (cousins, sexy pairs) exhibit ω-distortions reproduced by a two-centre Chinese-Remainder mechanism with no fitted parameters. We extend this from two members to three: the prime triplet $(6N-1, 6N+1, 6N+5)$, whose members sit on centre N (both wings) and its neighbour $N+1$ (left wing). Its dependence on $\omega_{>3}(N)$ is *non-monotone*—a rise to a peak near $\omega = 3$ followed by a fall—the signature of competing effects, twin-like enrichment on N versus the free neighbour $N+1$, exactly as for the sexy-B pair. The three-member two-centre CRT factor, with only the wing geometry as input, reproduces this shape on two shells: error $\leq 2.4\%$ for $\omega \leq 6$ on S_{10} ($\sim 2.3 \times 10^6$ triplets), with the $\omega = 6$ residual tightening from 6.5% (56 triplets on S_9) to 0.5% (1832 on S_{10}). The triplet is thus a three-member instance of the same two-centre mechanism. We note that the Hardy–Littlewood triplet constant $\mathfrak{S}(0, 2, 6) = 2.858$ is classical; the contribution is the non-monotone ω-shape and its two-centre CRT account. No claim is made about the infinitude of prime triplets.

1 The triplet straddles two centres

The only admissible triplet patterns within a span of six are $(0, 2, 6)$ and its mirror $(0, 4, 6)$; $(0, 2, 4)$ is inadmissible (it occupies all residues mod 3). On the $6N$ skeleton, $(p, p+2, p+6)$ with $p = 6N-1$ becomes

$$(\,6N-1,\ 6N+1,\ 6N+5\,) = (\,6N-1,\ 6N+1,\ 6(N+1)-1\,),$$

so its three members occupy, as (centre-offset, wing) pairs, $(0, -1), (0, +1), (1, -1)$: two wings on centre N and one on the neighbour $N+1$. Like the cousin and sexy pairs of Parts IX–X, and unlike the single-centre twin, the triplet *straddles two consecutive centres*. Its ω-dependence should therefore be a two-centre distortion, not the monotone single-centre enrichment of the twin.

2 A non-monotone distortion

Binning triplets by $\omega_{>3}(N)$ and normalising to $\omega = 1$, the rate is non-monotone: it rises to a peak near $\omega = 3$ and then falls (Table 1, Fig. 1). This is the sexy-B signature of Part IX: two competing effects. The twin component $(6N \pm 1)$ on the conditioned centre N enriches with ω (a factor-rich N protects both its wings), while the third member on the neighbour $N+1$ behaves like the cousin's partner (a factor-rich N does not protect a wing on $N+1$), suppressing the rate at high ω. The balance produces the peak.

3 The three-member two-centre CRT model

We extend the Part X factor to three members. For a prime $q > 3$, with $\mathrm{dead}(q, s) = -s\,6^{-1} \bmod q$ the residue of N making the wing $6(N + \mathrm{off}) + s$ divisible by q, the triplet's CRT factor is

$$f_q = \begin{cases} \prod_i \mathbf{1}\{\mathrm{off}_i \not\equiv \mathrm{dead}(q, s_i)\} & q \mid N, \\ \dfrac{\#\{r \neq 0 :\ \forall i,\ r + \mathrm{off}_i \not\equiv \mathrm{dead}(q, s_i)\}}{q - 1} & q \nmid N, \end{cases}$$

over the three members $\{(0, -1), (0, +1), (1, -1)\}$. The rate model is $\prod_{q \in \mathrm{POOL}} f_q$ evaluated per centre and averaged within each ω stratum; there are no fitted parameters.

Theorem 1 (three-member reproduction)**.** *On S_{10} ($\sim 2.3 \times 10^6$ triplets), the three-member two-centre CRT model reproduces the non-monotone ω-shape to $\leq 2.4\%$ for $\omega \leq 6$.*

S_9				S_{10}			
ω	triplets	meas	model	ω	triplets	meas	model
1	56114	1.000	1.000	1	356239	1.000	1.000
2	130928	1.034	1.024	2	875673	1.025	1.021
3	102726	1.055	1.039	3	768157	1.048	1.036
4	31012	1.020	1.006	4	288800	1.039	1.020
5	3071	0.870	0.852	5	43127	0.950	0.927
6	56	0.513	0.480	6	1832	0.691	0.687
				7	11	0.327	0.254

Table 1: Two-centre CRT model versus measurement, normalised to $\omega = 1$. The $\omega = 6$ residual falls from 6.5% (56 triplets, S_9) to 0.5% (1832, S_{10}): the S_9 excess was small-sample. The $\omega = 7$ stratum (11 triplets) is too small to read; the main result rests on $\omega \leq 6$, where the error is $\leq 2.4\%$.

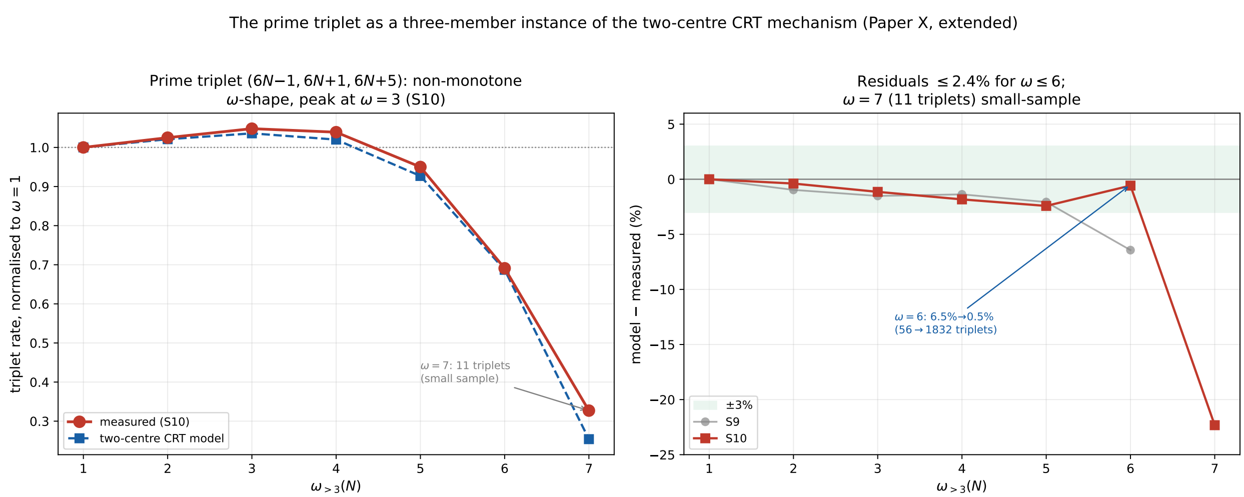

Figure 1: Left: the non-monotone triplet ω-shape (peak at $\omega = 3$), model versus measured on S_{10}. Right: residuals within $\pm 3\%$ for $\omega \leq 6$ on both shells; the $\omega = 6$ point tightens from S_9 to S_{10} as the sample grows; $\omega = 7$ (11 triplets) is small-sample.

The decisive check on the lone large S_9 residual is the shell comparison: at $\omega = 6$ the error falls from 6.5% to 0.5% as the count grows from 56 to 1832, identifying the S_9 excess as sampling noise. At $\omega = 7$ only 11 triplets exist on S_{10}; the 22% residual there is not interpretable and the result is stated for $\omega \leq 6$.

4 Scope and attribution

The triplet is a three-member instance of the two-centre mechanism of Parts IX–X: the same CRT factor, the same competition between the conditioned centre and its neighbour, now with three wings instead of two, and the same non-monotone outcome as sexy-B. The Hardy–Littlewood triplet singular series $\mathfrak{S}(0, 2, 6) = 2.858$ (which our sieve reproduces) is classical and sets the overall density; this paper does not propose a new constant. The contribution is the identification of the triplet's ω-dependence as a non-monotone two-centre distortion and its quantitative reproduction by the parameter-free CRT model. As throughout the series, no statement is made about the infinitude of prime triplets or any k-tuple conjecture; this is a measured, factor-resolved account of the triplet's conditional rate.

5 Conclusion

The prime triplet $(6N-1, 6N+1, 6N+5)$ carries a non-monotone ω-distortion—a peak at $\omega = 3$—reproduced to $\leq 2.4\%$ for $\omega \leq 6$ by the three-member extension of the two-centre CRT mechanism, with no fitted parameters and the $\omega = 6$ residual confirmed small-sample by the $S_9 \to S_{10}$ tightening. The triplet thus joins the cousin and sexy pairs as a two-centre pattern whose conditional shape is set by the competition between a factor-rich centre and its free neighbour, extending the mechanism from two members to three.

Data and code availability. The triplet counts, the two-centre CRT evaluation, and the stratified statistics are openly available [5].

References

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